Classical versus quantum entanglement entropy of coupled rotors: a universal deficit $1-\ln 2$ and its phase-space origin

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We compare the entanglement entropy of eigenstates of two rotors coupled by $\lambda\cos(\theta_1-\theta_2)$ — the phase model of two Josephson-coupled condensates, or two stacked planar dipolar rotors — with the Boltzmann entropy of the reduced classical microcanonical distribution and with the entropies of the Wigner and Husimi distributions. Conservation of total angular momentum makes the reduced density matrix diagonal and fixes the phase-space cell at h, so the comparison contains no adjustable constant. Every sector is an exact Mathieu problem. At weak coupling the $O(\lambda^2)$ normalization of the perturbed state supplies the entire linear entropy. Deep in the rotation regime the Schmidt weights are $J_j^2(\gamma)$, the classical weights follow the arcsine law, and the quantum entropy lies below the classical one by the universal constant $1-\ln 2$, approached as $\gamma^{-1/3}$. Both entropies peak near the separatrix. The Wigner function of the reduced state reproduces the von Neumann entropy identically; the constant $1-\ln 2$ is the entropy carried by the chord (interference) terms of the two-rotor Wigner function. The Husimi (Wehrl) entropy exceeds the von Neumann entropy by $(1-\ln 2)-D(\sigma)$ with $D(\sigma)\simeq 0.080/\sigma$ for coherent-state width σ: smoothing restores the classical value only as $\sigma\to\infty$, and the width takes over the role of the arbitrary phase-space cell of the coupled-oscillator problem.

I. INTRODUCTION

How much of the entanglement of a highly excited eigenstate can be obtained from a purely classical calculation? Duan, Kovner and Skokov (DKS) [1] posed this question for two coupled harmonic oscillators,

$$H_c=\tfrac{1}{2}\left(p_x^2+p_y^2+\omega^2x^2+\Omega^2y^2+2Cxy\right), \qquad (1)$$

with $C\ll\omega^2\ll\Omega^2$. They integrated the microcanonical distribution over the second oscillator, computed the Boltzmann entropy of the reduced distribution, and found it equal to the von Neumann entropy of the reduced density matrix of a WKB eigenstate, $S=\ln[\pi C\sqrt{E_1E_2}/(\hbar\omega\Omega^2)]$, provided the phase-space cell is taken to be $h/2$. Any other cell shifts the classical entropy by a constant. The question is motivated by high-energy scattering, where the entanglement of the probed degrees of freedom has been argued to appear as the Boltzmann entropy of the final state [2–4].

The oscillator pair is Gaussian, its phase space is noncompact (so the cell is not fixed by the theory), and it has no separatrix. Two rotors on a circle,

$$H=\frac{p_1^2}{2I}+\frac{p_2^2}{2I}+\lambda\cos\left(\theta_1-\theta_2\right), \qquad (2)$$

remove all three restrictions while remaining solvable. The model is realized in several settings. For two Bose–Einstein condensates coupled by tunneling, or two superconducting islands coupled by a Josephson junction, θ_i is the phase and $m_i=p_i/\hbar$ the particle number of mode i; in the Josephson regime, where the number imbalance is small compared with the total number, the two-mode Hamiltonian reduces to Eq. (2) with $\hbar^2/I$ the charging (interaction) energy and λ the Josephson energy [5–8]. The separatrix of the relative motion is then the boundary between Josephson oscillations and running-phase (self-trapped) motion [5, 6, 9], and the entanglement between the two rotors is the mode entanglement of the two condensates [10]. Two planar rotors carrying dipoles d, rotating in parallel planes a distance R apart along their common axis, interact through exactly $\lambda\cos(\theta_1-\theta_2)$ with $\lambda=d^2/4\pi\epsilon_0R^3$, because the dipoles are perpendicular to the separation.

We carry out the DKS comparison for Eq. (2) and extend it to the Wigner and Husimi distributions, whose information-theoretic content is of independent interest [11–13]. Our main results are as follows.

(i) Conservation of total angular momentum fixes the Schmidt basis and the phase-space cell, and every sector is an exact Mathieu problem (Sec. II).

(ii) At weak coupling the entropy is controlled by the $O(\lambda^2)$ normalization of the perturbed state, which supplies the whole linear entropy (Sec. III).

(iii) Deep in rotation the Schmidt weights are squared Bessel functions and the classical weights follow the arcsine law; the quantum entropy lies below the classical one by $1-\ln 2$ (Sec. V). At generic energies the deficit grows towards the same constant, and both entropies are maximal near the separatrix (Sec. VI). In deep libration we obtain the entropy including the first anharmonic correction (Sec. VII).

(iv) The reduced Wigner function returns the von Neumann entropy identically, and $1-\ln 2$ is the entropy of the chord terms of the Wigner function. The Wehrl entropy recovers a fraction $1-D(\sigma)/(1-\ln 2)$ of the deficit, with D in closed form (Sec. VIII).

Derivations are collected in the appendixes, and every analytic statement has been checked numerically (Appendix E).

II. MODEL AND EXACT REDUCTION

A. Conservation laws and relative motion

In the basis $\langle\theta_1,\theta_2|m_1,m_2\rangle = e^{im_1\theta_1+im_2\theta_2}/2\pi$, $m_i \in \mathbb{Z}$, the free energies are $\hbar^2(m_1^2+m_2^2)/2I$ and the coupling has matrix elements

$$\langle m_1', m_2'|\cos(\theta_1-\theta_2)\,|m_1,m_2\rangle = \tfrac{1}{2}\,\delta_{m_1',m_1\pm1}\,\delta_{m_2',m_2\mp1}, \tag{3}$$

transferring one quantum of angular momentum between the rotors. The total angular momentum $J = p_1 + p_2 = \hbar M$ is conserved. Within a sector we use

$$k = \frac{m_1-m_2}{2}, \qquad n = 2k, \qquad m_{1,2} = \frac{M}{2} \pm k, \tag{4}$$

with $k \in \mathbb{Z}$ for even M and $k \in \mathbb{Z}+\frac{1}{2}$ for odd M, and the dimensionless coupling $\alpha = \lambda I/2\hbar^2$.

Classically, the canonical variables $\Theta = (\theta_1+\theta_2)/2$, $\varphi = \theta_1-\theta_2$, J, and $p = (p_1-p_2)/2$ give

$$H = \frac{J^2}{4I} + H_{\rm rel}, \qquad H_{\rm rel} = \frac{p^2}{I} + \lambda\cos\varphi. \tag{5}$$

The relative motion is a pendulum with moment of inertia $\mu = I/2$, potential minimum at $\varphi = \pi$, and separatrix at $E_{\rm rel} \equiv E - J^2/4I = \lambda$; for $|E_{\rm rel}| < \lambda$ the relative angle librates, for $E_{\rm rel} > \lambda$ it circulates. Quantum mechanically $p = \hbar k$.

B. Each sector is a Mathieu problem

With $m_1^2 + m_2^2 = M^2/2 + 2k^2$, the Hamiltonian restricted to sector M is $\hbar^2M^2/4I + h$ with

$$h_{kk'} = \frac{\hbar^2}{I}k^2\,\delta_{kk'} + \frac{\lambda}{2}\left(\delta_{k',k+1} + \delta_{k',k-1}\right). \tag{6}$$

Writing $\Psi = e^{iM\Theta}u(\varphi)/\sqrt{2\pi}$ with $u = \sum_k c_k e^{ik\varphi}/\sqrt{2\pi}$, Eq. (6) is the matrix of $h = -(\hbar^2/I)\,\mathrm{d}^2/\mathrm{d}\varphi^2 + \lambda\cos\varphi$ in the plane-wave basis: each sector is exactly the quantum pendulum $H_{\rm rel}$, with periodic (M even) or antiperiodic (M odd) boundary conditions. With $\varphi = 2z$,

$$\begin{gathered} u'' + (a - 2q\cos 2z)u = 0, \\ a = \frac{4IE_{\rm rel}}{\hbar^2}, \qquad q = \frac{2\lambda I}{\hbar^2} = 4\alpha, \end{gathered} \tag{7}$$

the Mathieu equation [14]. The relative energies are $\hbar^2/4I$ times $a_0 < b_2 < a_2 < \ldots$ (M even) or $b_1 < a_1 < b_3 < \ldots$ (M odd), the separatrix is at $a = 2q$, and the Bohr–Sommerfeld area inside the separatrix, $8\sqrt{2I\lambda}$, gives $N_{\rm lib} \simeq (4/\pi)\sqrt{q}$ librating levels (25.5 and 50.9 for $\lambda I/\hbar^2 = 200$ and 800, against 25 and 51 counted).

C. The entanglement entropy is a Shannon entropy

An eigenstate in sector M is $|\Psi\rangle = \sum_k c_k \left|\frac{M}{2}+k, \frac{M}{2}-k\right\rangle$. Distinct k carry distinct rotor-2 labels, so the partial trace removes every cross term:

$$\begin{gathered} \rho_1 = \sum_k p_k \left|\tfrac{M}{2}+k\right\rangle\!\left\langle\tfrac{M}{2}+k\right|, \qquad p_k = |c_k|^2, \\ S_q = -\sum_k p_k \ln p_k, \end{gathered} \tag{8}$$

exactly. The entanglement entropy is the Shannon entropy of the Fourier coefficients of a Mathieu function, and the Schmidt basis is the angular-momentum basis at every energy and coupling. This is the structural fact on which everything below rests.

D. Exchange symmetry and branch states

The exchange $\theta_1 \leftrightarrow \theta_2$ maps $k \to -k$, so exact eigenstates have $c_{-k} = \pm c_k$. Above the separatrix the levels form doublets whose members are the even and odd combinations of two branch states $|\Psi_{\pm n}\rangle$, describing relative rotation in the two senses; the splitting vanishes as $\lambda^{|n|}$ at weak coupling and exponentially in the semiclassical regime. If the branch weights q_k are supported on $k > 0$, the parity eigenstate has $p_{\pm k} = q_k/2$ and

$$S_{\rm exact} = \ln 2 + S_{\rm branch}. \tag{9}$$

Since ρ_1 is diagonal, no cross term can contribute; only overlapping supports could. Classically the two branches are the two invariant tori $p \gtrless 0$, which the microcanonical distribution weights equally, so the same $\ln 2$ appears on both sides. Numerically we diagonalize Eq. (6) separately in the two parity sectors (Appendix E).

III. WEAK COUPLING AND THE NORMALIZATION TERM

Expand about $|0\rangle \equiv |m_1,m_2\rangle$ with $n = m_1 - m_2 \notin \{\pm1,\pm2\}$ (at $n = \pm1, \pm2$ the exchange partner is reached at first or second order). Rayleigh–Schrödinger theory in intermediate normalization (Appendix A) gives $E^{(1)} = 0$,

$$E^{(2)} = \frac{\lambda^2 I}{2\hbar^2\,(n^2-1)}, \tag{10}$$

first-order amplitudes $c_\pm = \mp\alpha/(n\pm1)$ on $|m_1\pm1, m_2\mp1\rangle$ and second-order amplitudes $c_{\pm\pm} = \alpha^2/[2(n\pm1)(n\pm2)]$ on $|m_1\pm2, m_2\mp2\rangle$. The normalization factor is $\mathcal{N} = 1 - \frac{1}{2}(\varepsilon_+ + \varepsilon_-) + O(\lambda^4)$ with

$$\varepsilon_\pm = c_\pm^2 = \frac{\alpha^2}{(n\pm1)^2}. \tag{11}$$

In the state, $\mathcal{N}-1$ is of the order to which the state is not kept and is routinely dropped. In the density matrix it cannot be: ρ is bilinear in the amplitudes, so the $|0\rangle\langle 0|$ element is $\mathcal{N}^2 = 1-\varepsilon_+-\varepsilon_-$, an $O(\lambda^2)$ correction of the same order as the populations $\varepsilon_\pm$. By Eq. (8) all coherences drop out of the partial trace (including the second-order amplitudes, which pair with $|0\rangle$ through different rotor-2 labels), and

$$\begin{aligned}\rho_1 = {} & (1-\varepsilon_+-\varepsilon_-)\,|m_1\rangle\langle m_1| \\ & + \varepsilon_+\,|m_1{+}1\rangle\langle m_1{+}1| + \varepsilon_-\,|m_1{-}1\rangle\langle m_1{-}1|\,,\end{aligned} \tag{12}$$

with error $O(\lambda^4)$. Hence

$$S = \varepsilon_+ \ln\frac{1}{\varepsilon_+} + \varepsilon_- \ln\frac{1}{\varepsilon_-} + (\varepsilon_+ + \varepsilon_-) + O(\varepsilon^2), \tag{13}$$

$$S_L = 1 - \operatorname{Tr}\rho_1^2 = 2\,(\varepsilon_+ + \varepsilon_-) + O(\varepsilon^2). \tag{14}$$

The non-logarithmic term of S and the whole of S_L come from the normalization: with $p_0 = 1$ one would find $\operatorname{Tr}\rho_1 = 1+\varepsilon_++\varepsilon_-$ and $S_L < 0$, and the von Neumann entropy would be low by a relative amount $1/[1+\ln(1/\varepsilon)]$, which vanishes only logarithmically as $\lambda \to 0$ (10–14% for $\lambda I/\hbar^2 = 0.02$–0.2 at $n = 0$, Fig. 1). The same structure appears in the oscillator ground state of Ref. [1], $S = -(1-f)\ln(1-f) - f\ln f$ with $f = C^2/4\omega\Omega^3$, where $-(1-f)\ln(1-f)$ is the normalization contribution. For $|n| \geq 3$ the exact parity eigenstates have entropy $\ln 2$ plus Eq. (13), by Eq. (9), since the two branches overlap only at $O(\lambda^{|n|})$.

For $|n| \gg 1$, $\varepsilon_+ \simeq \varepsilon_- \simeq \gamma^2/4$ with

$$\gamma \equiv \frac{\lambda I}{\hbar^2 n} = \frac{2\alpha}{n}, \qquad S \simeq \gamma^2\left[\ln\frac{2}{\gamma} + \frac{1}{2}\right]. \tag{15}$$

The factor 2 in the logarithm is easily lost; with it, Eq. (15) agrees with Eq. (13) to 0.3% at $n = 30$. We shall recover Eq. (15) as the small-γ limit of an exact solution.

IV. THE CLASSICAL REDUCED DISTRIBUTION

Following Ref. [1], the classical counterpart of an eigenstate with energy E and total angular momentum J is the microcanonical distribution

$$W_{\rm cl} = \mathcal{C}\,\delta(E-H)\,\delta(J-p_1-p_2). \tag{16}$$

Integrating over (θ_2, p_2), the second delta function sets $p_2 = J - p_1$, and the first has two roots $\varphi = \pm\arccos\xi$, $\xi = (E_{\rm rel} - p^2/I)/\lambda$, with $p = p_1 - J/2$. The reduced distribution is

$$W_{\rm cl}(\theta_1, p_1) = \frac{2\mathcal{C}\;\Theta\big(\lambda - |E_{\rm rel} - p^2/I|\big)}{\sqrt{\lambda^2 - (E_{\rm rel} - p^2/I)^2}}, \tag{17}$$

independent of θ_1. The two roots $\pm\arccos\xi$ — two relative angles compatible with one momentum of rotor 1 — are the two branches whose quantum interference is the subject of Secs. V and VIII.

The support is $|p| \leq \sqrt{I(E_{\rm rel}+\lambda)}$ in libration and $\sqrt{I(E_{\rm rel}-\lambda)} \leq |p| \leq \sqrt{I(E_{\rm rel}+\lambda)}$ in rotation. Its length $L(E_{\rm rel})$ grows through libration, peaks at $2\sqrt{2I\lambda}$ at the separatrix, and falls above it with infinite slope: a cusp maximum, which bounds the entropy by $\ln(L/\hbar)$. The momentum density $w = 2\pi W_{\rm cl}$ is normalized by $\mathcal{C} = 1/8\pi B$, with the band integral (Appendix B)

$$B = \begin{cases} \sqrt{\dfrac{I}{E_{\rm rel}+\lambda}}\,\mathsf{K}\!\left(\sqrt{\dfrac{2\lambda}{E_{\rm rel}+\lambda}}\right), & E_{\rm rel} > \lambda, \\[2ex] \sqrt{\dfrac{I}{2\lambda}}\,\mathsf{K}\!\left(\sqrt{\dfrac{E_{\rm rel}+\lambda}{2\lambda}}\right), & |E_{\rm rel}| < \lambda, \end{cases} \tag{18}$$

where K is the complete elliptic integral of the first kind; $2B$ is the period of the relative motion, so $w\,\mathrm{d}p_1$ is the fraction of the period spent in $\mathrm{d}p_1$.

The cell. The Boltzmann entropy $-\int W_{\rm cl}\ln(W_{\rm cl}\Delta)$ requires a cell Δ. On a compact configuration space it is fixed: $p_1 = \hbar m_1$, and each quantum state occupies the strip $2\pi \times \hbar = h$. With $\Delta = h$, $S_{\rm cl} = -\int \mathrm{d}p_1\, w\ln(w\hbar)$. The quantity with the same status as Eq. (8) is the Shannon entropy $S_{\rm cl}^{\rm bin}$ of the cell weights $w_m = \int_{\hbar(m-1/2)}^{\hbar(m+1/2)} w\,\mathrm{d}p_1$. One has $S_{\rm cl}^{\rm bin} \geq S_{\rm cl}$, the difference being a sum of relative entropies concentrated in the cells that contain the band-edge singularities. Neither comparison contains an adjustable constant.

V. DEEP ROTATION AND THE CONSTANT $1-\ln 2$

A. Wannier–Stark ladder and Bessel weights

For $k = n/2 + j$ with $n \gg 1$, $\hbar^2k^2/I = \hbar^2n^2/4I + Fj + \hbar^2j^2/I$ with $F = \hbar^2 n/I$. Dropping the quadratic term, which is smaller than the linear one by $|j|/n$, turns Eq. (6) into the Wannier–Stark problem $Fjc_j + \frac{\lambda}{2}(c_{j+1} + c_{j-1}) = Ec_j$ [15, 16]. The Bessel recurrence $J_{\nu-1} + J_{\nu+1} = (2\nu/x)J_\nu$ shows that $c_j = J_{j-s}(-\lambda/F)$ is an eigenvector with eigenvalue Fs, so

$$p_j = J_{j-s}^2(\gamma), \qquad \gamma = \frac{\lambda}{F} = \frac{\lambda I}{\hbar^2 n}, \tag{19}$$

valid for $\gamma/n = \lambda/4E_{\rm rel} \ll 1$. Exact diagonalization confirms Eq. (19) pointwise to 3×10^{-3} at $\gamma/n = 0.01$, and the entropy to four or five digits for $n = 100$–400. The Bessel eigenvector is a branch state, and its entropy

$$S_{\rm branch}(\gamma) = -\sum_j J_j^2(\gamma)\ln J_j^2(\gamma) \tag{20}$$

is a universal function of γ (Fig. 2). For small γ, $J_0^2 \simeq 1-\gamma^2/2$ and $J_{\pm1}^2 \simeq \gamma^2/4$ reproduce Eq. (15), the depletion of J_0^2 being the normalization term of Sec. III.

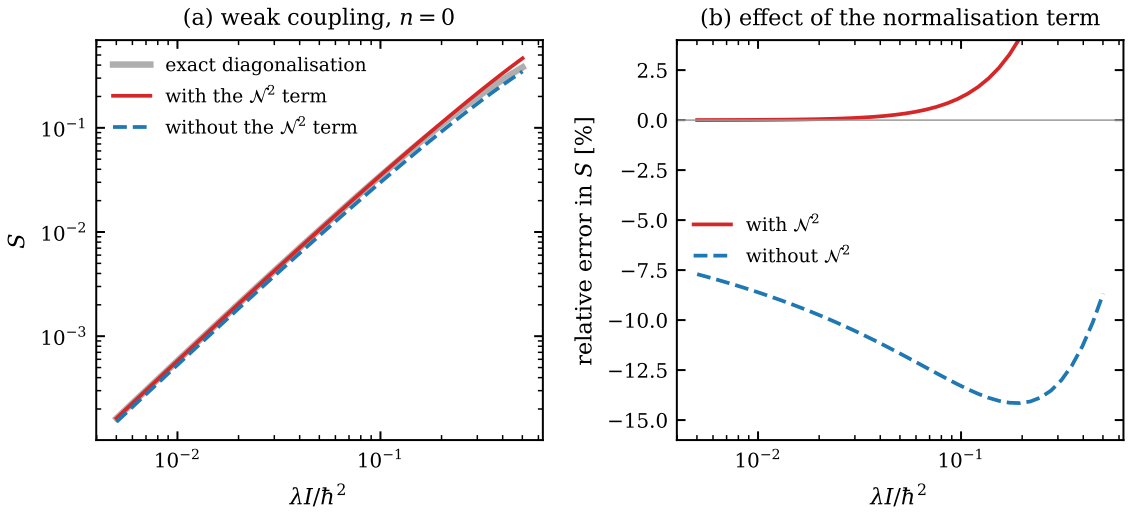


FIG. 1. Weak coupling, $n = 0$ ($m_1 = m_2 = 4$). (a) Entanglement entropy from exact diagonalization (grey), Eq. (13) (red), and the same expression with the normalization term removed (blue dashed). (b) Relative error of the two truncations: with the term, 0.05% at $\lambda I/\hbar^2 = 0.02$ and 1% at 0.1, decreasing as λ^2; without it, 10–14%, not vanishing as $\lambda \to 0$.

In the same limit the classical density (17), linearized about $p_0 = \hbar n/2$, becomes the arcsine law of half-width γ,

$$w(j) = \frac{1}{\pi\sqrt{\gamma^2 - j^2}}, \qquad S_{\rm cl}^{\rm branch} = \ln\frac{\pi\gamma}{2}, \tag{21}$$

so that for the two bands of a parity eigenstate $S_{\rm cl} = \ln(\pi\gamma) = \ln[(\pi\lambda/2\hbar)\sqrt{I/E_{\rm rel}}]$, the rotor counterpart of the DKS result.

B. Fringes and the entropy deficit

The Debye expansion of $J_j(\gamma)$ with $j = \gamma\cos\beta$ [17] gives

$$\begin{aligned} J_j^2(\gamma) &\simeq w(j)\,[1+\cos\Phi_j]\,, \\ \Phi_j &= 2\sqrt{\gamma^2 - j^2} - 2j\arccos\frac{j}{\gamma} - \frac{\pi}{2}, \end{aligned} \tag{22}$$

fringes of *full* contrast about the classical weights, with local wavenumber $\mathrm{d}\Phi_j/\mathrm{d}j = -2\beta \in (-2\pi, 0)$. The same holds at any energy: the WKB ansatz $c_k = A_k e^{i\chi_k}$ for the difference equation $\hbar^2k^2c_k/I + \frac{\lambda}{2}(c_{k+1} + c_{k-1}) = E_{\rm rel}c_k$ gives a local wavenumber ϕ_k with $E_{\rm rel} - \hbar^2k^2/I = \lambda\cos\phi_k$ — the classical relative angle — and constancy of the Casoratian gives $|A_k|^2 \propto [\lambda^2 - (E_{\rm rel} - \hbar^2k^2/I)^2]^{-1/2}$, the classical density. A real eigenvector superposes the two waves $\pm\phi_k$, so

$$p_k \simeq w_k\,(1+\cos\Phi_k)\,, \qquad \Phi_{k+1} - \Phi_k = 2\phi_k. \tag{23}$$

Inserting this into Eq. (8), the cross term $\sum w_k\cos\Phi_k\ln w_k$ averages out and the phases are equidistributed wherever $2\phi_k \not\approx 0 \bmod 2\pi$, so

$$S_{\rm cl} - S_q \;\to\; \frac{1}{2\pi}\int_0^{2\pi}(1+\cos\Phi)\ln(1+\cos\Phi)\,\mathrm{d}\Phi = 1 - \ln 2 \tag{24}$$

(Appendix B). The limiting deficit $1 - \ln 2 = 0.30685$ is universal: it depends only on the full contrast of the two-branch interference. Equidistribution fails near the band edges (caustics), where the fringes become stationary; the phase accumulated over the last δ cells, $\simeq \frac{4\sqrt{2}}{3}\delta^{3/2}/\sqrt{\gamma}$, reaches 2π at $\delta \sim \gamma^{1/3}$, whose arcsine weight is $\sim \gamma^{-1/3}$. Hence $S_{\rm cl} - S_q = 1 - \ln 2 - O(\gamma^{-1/3})$. Summing the Bessel weights directly, $S_{\rm cl}^{\rm bin} - S_q = 0.233, 0.270, 0.288$ and 0.298 at $\gamma = 10^3, 10^4, 10^5, 10^6$, the distance to $1 - \ln 2$ falling by close to $10^{1/3}$ per decade [Fig. 2(b)].

Why oscillators and rotors differ. For the oscillators no conservation law selects a Schmidt basis; the reduced density matrix of Ref. [1] is doubly degenerate with smooth arcsine eigenvalues, and the two-branch interference is absorbed into the eigenvectors. For the rotors, conservation of J fixes the Schmidt basis (Sec. II C), so the interference appears in the eigenvalues as the fringes (23). The leading logarithms agree; the constants differ by $1 - \ln 2$. Equivalently, the classical formula reproduces the quantum entropy if the cell is taken to be $eh/2$ instead of h.

VI. GENERIC ENERGIES AND THE SEPARATRIX

Figure 3 compares S_q from exact diagonalization with $S_{\rm cl}^{\rm bin}$ from quadrature of Eq. (17) for $\lambda I/\hbar^2 = 50$, 200 and 800. The curves track each other over the whole spectrum. In libration the quantum entropy lies below

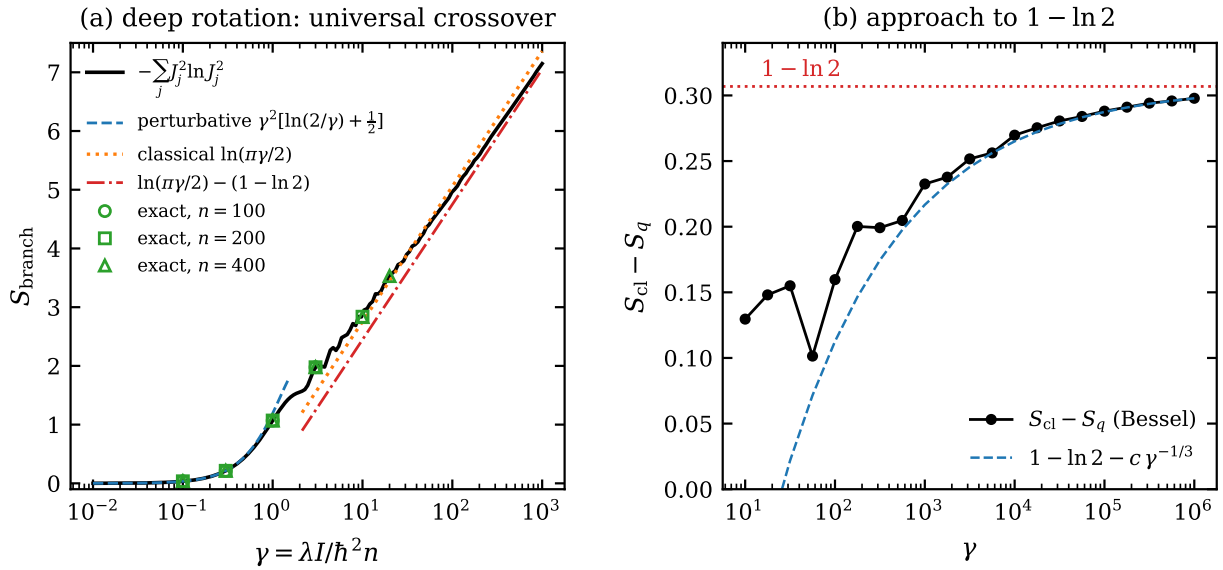


FIG. 2. Deep rotation. (a) Branch entropy as a function of $\gamma = \lambda I/\hbar^2 n$: Bessel result (20) (black), its small-γ limit (15) (blue dashed), classical arcsine entropy (orange dotted), and classical entropy minus $1-\ln 2$ (red dash-dotted). Open symbols: exact diagonalization at $n = 100, 200, 400$. (b) Deficit $S_{\rm cl}^{\rm bin} - S_q$ from the Bessel weights, approaching $1-\ln 2$ as $\gamma^{-1/3}$.

the classical one by about 0.2 (except near the bottom of the well, where the band covers only a few cells); in rotation by a smaller amount that grows with λ. Figure 4(a) shows the mechanism at $\lambda I/\hbar^2 = 800$, $E_{\rm rel}/\lambda = 1.54$: the quantum weights oscillate about the classical envelope with full contrast, and although the total-variation distance between the two distributions is 0.33, the entropies differ by less than 1%. Averaged over $0.2 \le E_{\rm rel}/\lambda \le 0.7$ and $1.5 \le E_{\rm rel}/\lambda \le 2.5$, the deficit grows from 0.18 to 0.24 (libration) and from 0.01 to 0.15 (rotation) as $\lambda I/\hbar^2$ increases from 200 to 51 200 [Fig. 4(b)], monotonically towards $1-\ln 2$ but more slowly than in deep rotation, because the bands contain only 10–500 cells and the caustic regions carry a large share of the weight.

Both entropies are maximal near the separatrix, at $E_{\rm rel}/\lambda = 1.13$, 1.12 and 0.90 for the three couplings, because the support of the classical momentum distribution has its cusp maximum there. The deviation in Fig. 3(b) changes abruptly at the separatrix because the fringe structure changes: below it the interfering branches are the two halves of a libration orbit, above it of a rotation orbit, and new caustics appear at $p = \pm\sqrt{I(E_{\rm rel}-\lambda)}$. In the Josephson-junction realization this is the boundary between Josephson oscillations and running-phase motion [5, 6]; more generally the separatrix is the classical locus of an excited-state quantum phase transition [18–20].

VII. DEEP LIBRATION

For $\lambda I/\hbar^2 \gg 1$ the ground state of h is localized at $\varphi = \pi$. With $x = \varphi - \pi$, $h = -\lambda + p^2/2\mu + \frac{1}{2}\mu\omega^2 x^2 - g x^4 + \ldots$ with $\mu = I/2$, $\omega = \sqrt{2\lambda/I}$ and $g = \lambda/24$. In the harmonic approximation the Fourier coefficients form a Gaussian in k of variance $\sigma_0^2 = \mu\omega/2\hbar = \frac{1}{4}\sqrt{2\lambda I/\hbar^2}$, and by Poisson summation the Shannon entropy of a sampled Gaussian is $\frac{1}{2}\ln(2\pi e\sigma^2)$ up to $O(\sigma^2 e^{-2\pi^2\sigma^2})$. The quartic term narrows the distribution: the virial theorem gives $\langle p^2\rangle = \mu\hbar\omega/2 - \hbar^2/16$ to first order in g, and the non-Gaussian distortion does not change the entropy at first order because $\ln\phi_\sigma$ is quadratic in k (Appendix C). Hence

$$S_0 = \frac{1}{4}\ln\frac{\lambda I}{\hbar^2} + \frac{1}{2}\ln\frac{\pi e}{\sqrt{2}} - \frac{1}{8\sqrt{2\lambda I/\hbar^2}} + O\!\left(\frac{\hbar^2}{\lambda I}\right). \quad (25)$$

For $\lambda I/\hbar^2 = 10$, 40, 160, 640, 2560, 10240 the ratio of the numerical residual (after the harmonic terms) to the predicted correction is 1.083, 1.039, 1.019, 1.009, 1.005, 1.002 (Fig. 5).

VIII. WIGNER AND HUSIMI DISTRIBUTIONS

We now replace the classical distribution (16) by a quantum phase-space distribution of the eigenstate. Momenta are in units of $\hbar$.

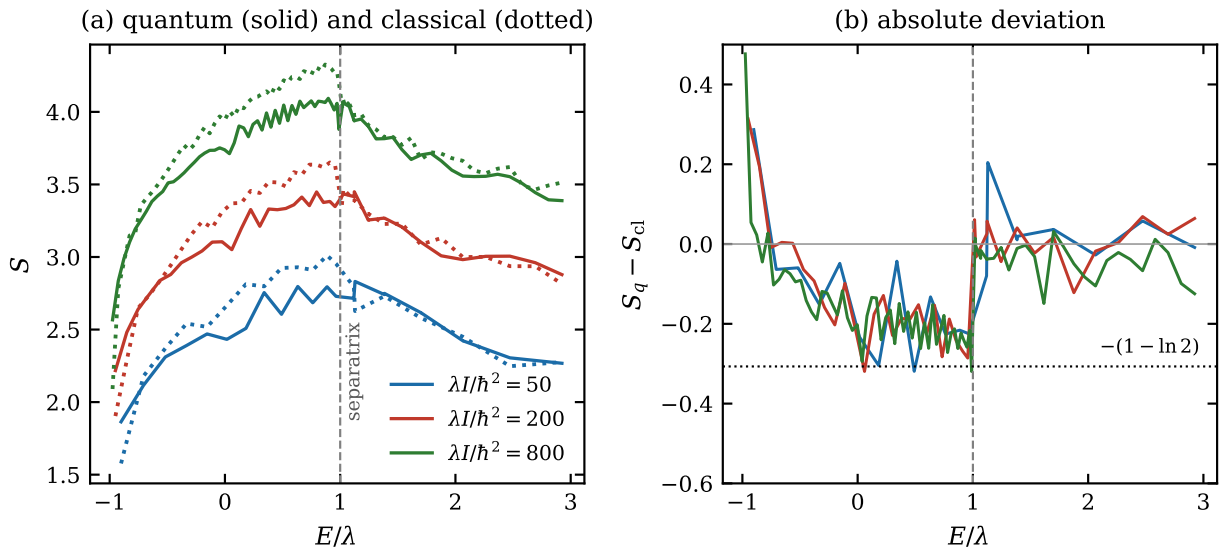


FIG. 3. (a) Quantum (solid) and classical coarse-grained (dotted) entropies versus $E_{\rm rel}/\lambda$ for $\lambda I/\hbar^2 = 50$, 200, 800; dashed line: separatrix. (b) $S_q - S_{\rm cl}^{\rm bin}$, with the asymptotic value $-(1 - \ln 2)$ dotted.

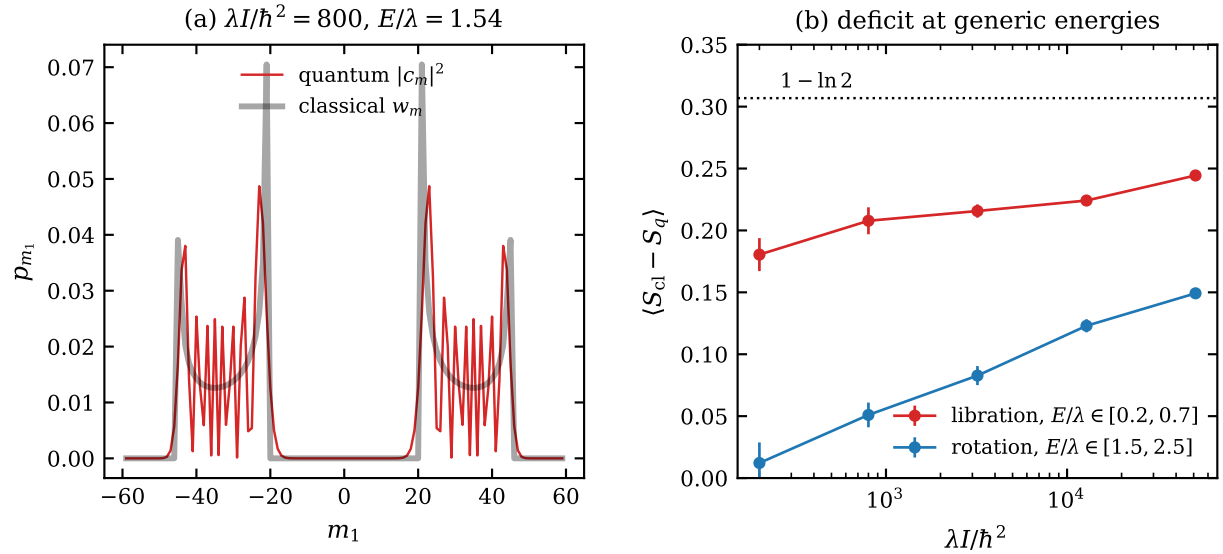


FIG. 4. (a) Momentum distribution of rotor 1 at $\lambda I/\hbar^2 = 800$, $E_{\rm rel}/\lambda = 1.54$: quantum weights (red) oscillate with full contrast about the classical cell weights (grey), as Eq. (23) predicts. (b) Deficit $\langle S_{\rm cl}^{\rm bin} - S_q \rangle$ averaged over energy windows in libration and rotation versus λ; error bars are standard errors over the window.

A. Wigner function: the DKS construction returns S_q

For a rotor the Wigner function lives on $\theta \in [0, 2\pi)$ and $\ell \in \frac{1}{2}\mathbb{Z}$ [21–23],

$$W_\rho(\theta, \ell) = \frac{1}{4\pi} \int_{-2\pi}^{2\pi} \mathrm{d}\xi \, e^{-i\ell\xi} \left\langle \theta + \tfrac{\xi}{2} \right| \rho \left| \theta - \tfrac{\xi}{2} \right\rangle = \frac{1}{2\pi} \sum_{m+m'=2\ell} \rho_{mm'} \, e^{i(m-m')\theta} \tag{26}$$

(Appendix D). It is real; $\sum_\ell W_\rho = \langle\theta|\,\rho\,|\theta\rangle$; $\int \mathrm{d}\theta\, W_\rho = \rho_{\ell\ell}$ for integer ℓ and 0 for half-integer ℓ; and $\mathrm{Tr}\,\rho\sigma = 2\pi \sum_\ell \int \mathrm{d}\theta\, W_\rho W_\sigma$. For two rotors, integrating the two-rotor Wigner function over (θ_2, ℓ_2) is the partial trace: $\sum_{\ell_2} \int \mathrm{d}\theta_2\, W = W_{\rho_1}$. For an eigenstate,

$$W(\theta_1, \ell_1; \theta_2, \ell_2) = \frac{1}{2\pi}\, \delta_{\ell_1+\ell_2, M}\, W_{\rm rel}\big(\theta_1 - \theta_2,\, \ell_1 - \tfrac{M}{2}\big), \tag{27}$$

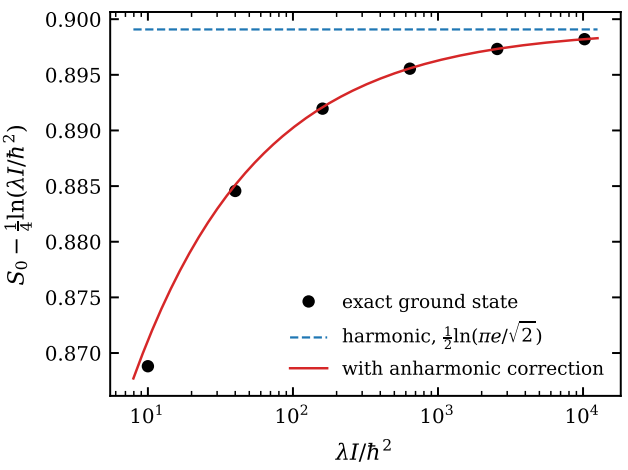


FIG. 5. Deep libration: $S_0 - \frac{1}{4}\ln(\lambda I/\hbar^2)$ for the ground state (points), the harmonic constant $\frac{1}{2}\ln(\pi e/\sqrt{2})$ (dashed), and Eq. (25) (solid).

with $W_{\rm rel}(\varphi,\ell) = (2\pi)^{-1}\sum_{k+k'=2\ell} c_k \bar{c}_{k'} e^{i(k-k')\varphi}$ the Wigner function of $u(\varphi)$. The microcanonical distribution has the same skeleton, $W_{\rm cl} = (2\pi)^{-1}\delta(J - p_1 - p_2)\,\delta(E_{\rm rel} - H_{\rm rel})/4B$: the only difference is $W_{\rm rel}$ versus the microcanonical density on the pendulum level set.

Carrying out the DKS construction with W in place of $W_{\rm cl}$, the reduced Wigner function is $W_1 = p_k/2\pi$ at $\ell_1 = \frac{M}{2} + k$ and zero on the other half-integer rows. It is non-negative (because ρ_1 is diagonal) and its Boltzmann entropy with cell h is

$$-\sum_{\ell_1}\int_0^{2\pi} \mathrm{d}\theta_1\, W_1 \ln(2\pi W_1) = -\sum_k p_k \ln p_k = S_q, \quad (28)$$

identically, for every eigenstate and coupling. Replacing the classical distribution by the Wigner function gives the quantum answer, not a classical one.

B. The deficit lives in the chord terms

Figure 6 shows $W_{\rm rel}$ for a librating and a rotating level. As for any one-dimensional eigenstate [24, 25], $W_{\rm rel}$ is large on the classical curve, with an Airy profile across it, and oscillates inside: a point (φ,ℓ) receives a contribution from each chord of the curve centered on it, with phase equal to the enclosed symplectic area over $\hbar$. The negativity volume [26] $\sum_\ell \int |W_{\rm rel}| - 1$ is 5.86 and 5.43 for the two states shown. Averaged over regions large compared with $\hbar$ but classically small, the chord terms cancel and $W_{\rm rel}$ reduces to the microcanonical density [24, 27].

Integrating $W_{\rm rel}$ over φ at integer ℓ collects (a) the two points of the classical curve at momentum ℓ, $\varphi = \pm\phi_\ell$, which give the classical weight w_ℓ, and (b) the horizontal chord joining them, centered on $\varphi = 0$ and π, which gives $w_\ell \cos\Phi_\ell$. At $\ell = 14$ in Fig. 6(c), windows of half-width 0.35 around $\pm\phi_\ell$ contribute $+0.022$, around 0 and π -0.027, and the total is $p_{14} = 7\times10^{-4}$: a dark fringe. On half-integer rows the chord contributions at 0 and π cancel (∓ 0.012 at $\ell = 16.5$). Smoothing before marginalizing gives the classical distribution; marginalizing first keeps the chord terms. *The constant $1 - \ln 2$ is the entropy carried by the chord terms of the Wigner function.*

C. Husimi function and Wehrl entropy

The Husimi function removes chord terms by smoothing over a coherent state. On the circle we use [28, 29]

$$|\theta,\ell\rangle = \sum_m \sqrt{g_\sigma(m-\ell)}\, e^{-im\theta}\,|m\rangle\,, \quad g_\sigma(x) = \frac{e^{-x^2/2\sigma^2}}{\sqrt{2\pi}\,\sigma}, \quad (29)$$

with $\ell \in \mathbb{R}$, momentum width σ and angular width $1/2\sigma$. The resolution of identity $\int \frac{\mathrm{d}\theta}{2\pi}\mathrm{d}\ell\, |\theta,\ell\rangle\langle\theta,\ell| = \mathbb{1}$ holds exactly with cell h, and the norm is 1 up to $2e^{-2\pi^2\sigma^2}$ (1.4% at $\sigma = \frac{1}{2}$). For the state (8) the Husimi function is $Q(\ell) = \sum_m p_m g_\sigma(\ell - m)$, independent of θ, and the Wehrl entropy [11, 30] is $S_W = -\int \mathrm{d}\ell\, Q \ln Q$. Applied to the classical density, the same smoothing gives $S_W[w] \to S_{\rm cl}$ as $\sigma \to 0$. Jensen's inequality (the Berezin–Lieb bound [31, 32]) and $h(M+Z) \le H(M) + h(Z)$ [33] give

$$S_q - 2e^{-2\pi^2\sigma^2} \lesssim S_W \le S_q + \tfrac{1}{2}\ln(2\pi e\sigma^2). \quad (30)$$

Deep rotation in closed form. Over a stretch of cells where w is nearly constant and $\Phi_j \simeq \Phi_0 + \kappa j$ with $\kappa = -2\beta$, Poisson summation gives $\sum_j e^{i\kappa j} g_\sigma(x-j) = \sum_r e^{i(\kappa+2\pi r)x} e^{-\sigma^2(\kappa+2\pi r)^2/2}$. For $\sigma \gtrsim 1$ only the alias nearest zero survives, and

$$Q \simeq w\left[1 + A\cos\tilde{\Phi}\right], \qquad A = e^{-\sigma^2 d^2/2}, \qquad d = \min(2\beta, 2\pi - 2\beta). \quad (31)$$

Smoothing reduces the fringe contrast from 1 to A without removing the fringes. Averaging over the fast phase, $S_W \simeq S_{\rm cl} - \int w\, f(A)$, with the contrast function (Appendix B)

$$f(A) = \big\langle (1 + A\cos\Phi)\ln(1 + A\cos\Phi)\big\rangle = \ln\frac{1+\sqrt{1-A^2}}{2} + 1 - \sqrt{1-A^2}, \quad (32)$$

which gives $f(1) = 1 - \ln 2$. Since the arcsine weight is uniform in β,

$$S_W - S_q = (1 - \ln 2) - D(\sigma), \qquad D(\sigma) = \frac{2}{\pi}\int_0^{\pi/2} f\big(e^{-2\sigma^2\beta^2}\big)\,\mathrm{d}\beta. \quad (33)$$

Both Eq. (23) and Eq. (31) fail near the caustics, but there $A \to 1$ and the integrand $f(1) - f(A)$ of $S_W - S_q$

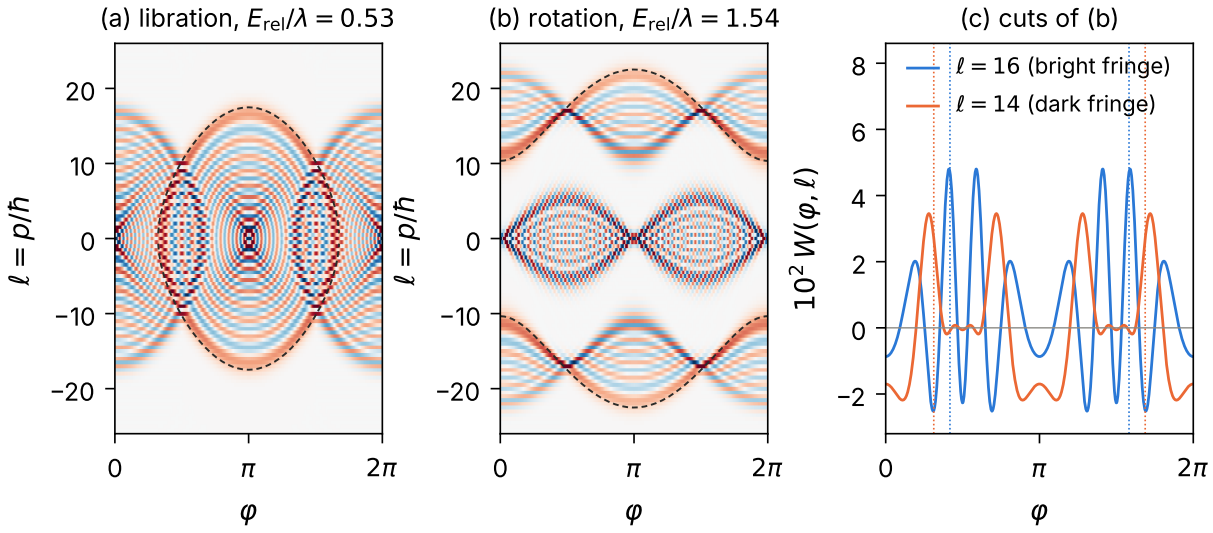


FIG. 6. Relative Wigner function $W_{\rm rel}(\varphi, \ell)$ at $\lambda I/\hbar^2 = 200$ (red positive, blue negative; integer and half-integer rows). (a) Librating level, $E_{\rm rel}/\lambda = 0.53$. (b) Rotating level, $E_{\rm rel}/\lambda = 1.54$, odd parity; the band near $\ell = 0$ is the interference of the two counter-rotating tori. Dashed: classical curve. (c) Cuts of (b) at $\ell = 16$ (bright fringe of p_ℓ) and $\ell = 14$ (dark fringe); dotted lines: classical angles $\pm\phi_\ell$. The oscillations away from them are the chord terms that survive integration over φ.

vanishes, so Eq. (33) holds without the $\gamma^{-1/3}$ correction. For $\sigma \gg 1$, expanding f in powers of A^2,

$$D(\sigma) \simeq \frac{c_1}{\sigma}, \qquad c_1 = \frac{1}{2\sqrt{\pi}} \sum_{n\geq 1} \frac{\binom{2n}{n}}{4^n\, 2n(2n-1)\sqrt{n}} = 0.080171, \tag{34}$$

accurate to 10^{-5} for $\sigma \geq 1$. Table I tests Eq. (33) (column "pred.") against the Bessel weights. At $\gamma = 10^4$ the agreement is within 0.001 for $1 \leq \sigma \leq 4$, and $S_W - S_q$ is the same at $\gamma = 10^3$ and 10^4 to 0.004 although $S_{\rm cl} - S_q$ changes from 0.212 to 0.263. At $\sigma = \frac{1}{2}$ a second alias lowers the result by 0.003; at $\gamma \leq 100$ the larger widths resolve the band itself.

Thus, at fixed σ and $\gamma \to \infty$,

$$S_W - S_{\rm cl} \;\to\; -D(\sigma) \simeq -0.080/\sigma : \tag{35}$$

the Husimi entropy does not converge to the classical one. The classical value is reached only for $1 \ll \sigma \ll \gamma$, and only algebraically, because fringes of every wavenumber down to zero exist near the band edges and a Gaussian of any width leaves the slowest intact [Fig. 7(a)].

Generic energies. The local fringe wavenumber is $2\phi_k$, and the same argument gives $S_W - S_q \simeq (1 - \ln 2) - \langle f(e^{-\sigma^2 d(2\phi)^2/2})\rangle_{\rm orbit}$, which agrees with the numerics to within 0.03 for $\sigma \leq 1$ in the windows of Sec. VI (Table II). The fair comparison smooths quantum and classical distributions with the same coherent state. For $\sigma = 1$ the two Wehrl entropies then agree to within 0.04, and for $\sigma = 2$ to within 0.015, at every level with $-0.5 < E_{\rm rel}/\lambda < 0.8$ or $1.2 < E_{\rm rel}/\lambda < 3$ at $\lambda I/\hbar^2 = 200$ and 800 [Fig. 7(b)], compared with an unsmoothed deficit of about 0.2 in libration. But the agreement is achieved by moving both entropies: $S_W[w]$ exceeds $S_{\rm cl}$ by 0.1–0.7, depending on σ and the width of the band.

TABLE I. Wehrl entropy of the deep-rotation weights $J_j^2(\gamma)$ against Eq. (33); $S_W[w]$ is the arcsine law smoothed with the same σ, and $S_{\rm cl} = \ln(\pi\gamma/2)$.

γ	σ	$S_W - S_q$	pred.	$S_W - S_{\rm cl}$	$S_W[w] - S_W$
10^2	0.5	0.1406	0.1502	0.0453	0.0364
	1	0.2218	0.2267	0.1265	−0.0110
	2	0.2713	0.2668	0.1760	−0.0127
	4	0.3300	0.2868	0.2347	−0.0040
10^3	0.5	0.1468	0.1502	−0.0654	0.0912
	1	0.2274	0.2267	0.0152	0.0213
	2	0.2680	0.2668	0.0558	−0.0041
	4	0.2916	0.2868	0.0795	−0.0064
10^4	0.5	0.1468	0.1502	−0.1165	0.1247
	1	0.2274	0.2267	−0.0359	0.0475
	2	0.2675	0.2668	0.0042	0.0121
	4	0.2877	0.2868	0.0244	−0.0013

The coherent-state width is not fixed by the problem. It therefore takes over the role of the cell Δ in the oscillator problem: a parameter outside the classical theory whose choice decides the constant term. The cell itself stays fixed at h by the resolution of identity. The width $\sigma_0 \propto (\lambda I/\hbar^2)^{1/4}$ of the pendulum ground state is a natural choice, for which $D(\sigma_0) \to 0$ in the semiclassical limit, but only as $(\lambda I/\hbar^2)^{-1/4}$, and the choice is a convention.

TABLE II. Window averages ($0.2 \le E_{\rm rel}/\lambda \le 0.7$ libration, $1.5 \le E_{\rm rel}/\lambda \le 2.5$ rotation), all levels of sector $M = 0$.

$\lambda I/\hbar^2$	regime	$S_{\rm cl}^{\rm bin} - S_q$	σ	$S_W - S_q$	$(1-\ln 2) - \langle f\rangle_{\rm orbit}$	$S_W[w] - S_W$	$S_W[w] - S_{\rm cl}$
200	libration	0.180	0.5	0.193	0.204	0.025	0.147
			1	0.284	0.270	−0.007	0.207
			2	0.364	0.289	−0.002	0.292
200	rotation	0.012	0.5	0.146	0.146	−0.046	0.361
			1	0.273	0.223	−0.026	0.508
			2	0.459	0.264	−0.007	0.713
800	libration	0.208	0.5	0.202	0.204	0.031	0.104
			1	0.284	0.270	−0.008	0.147
			2	0.339	0.289	−0.003	0.207
800	rotation	0.051	0.5	0.144	0.146	−0.030	0.256
			1	0.246	0.223	−0.027	0.361
			2	0.374	0.264	−0.009	0.508

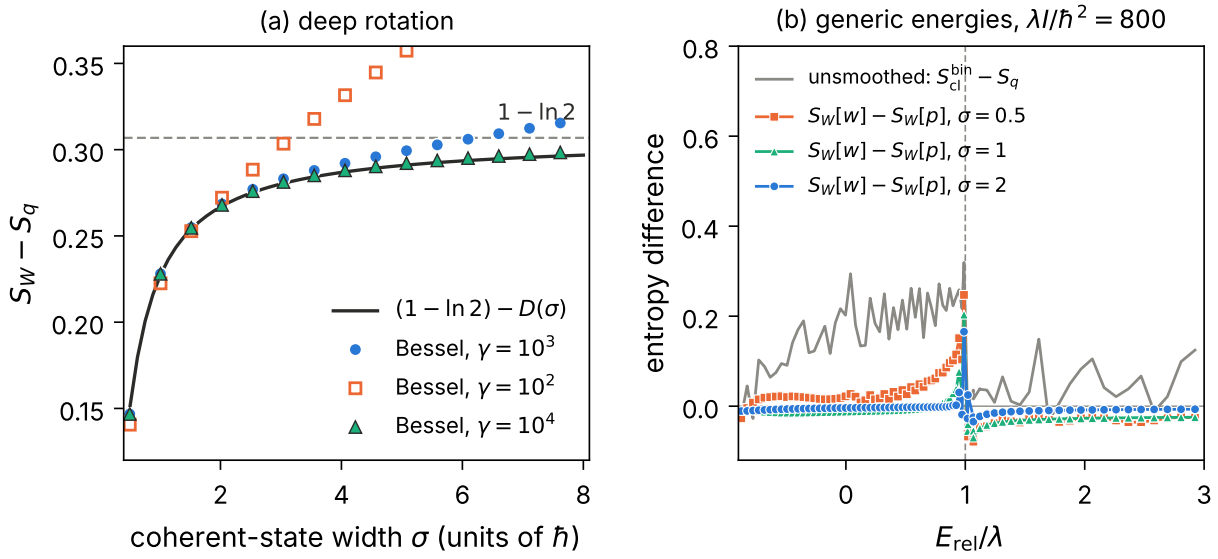


FIG. 7. (a) Deep rotation: excess of the Wehrl over the von Neumann entropy versus coherent-state width, Eq. (33) (solid), and Bessel weights at $\gamma = 10^2, 10^3, 10^4$ (symbols). The full deficit $1 - \ln 2$ (dashed) is recovered only as $\sigma \to \infty$. (b) Generic energies at $\lambda I/\hbar^2 = 800$: difference between the Wehrl entropies of the classical and quantum momentum distributions for $\sigma = 0.5, 1, 2$, and the unsmoothed difference $S_{\rm cl}^{\rm bin} - S_q$ (grey). Dashed line: separatrix.

IX. CONCLUSIONS

For two coupled rotors, conservation of total angular momentum fixes both the Schmidt basis and the phase-space cell, so the classical–quantum comparison of entanglement entropy proposed in Ref. [1] can be made with nothing to adjust. The leading logarithms agree, as for oscillators, but the quantum entropy lies below the classical one by the universal constant $1 - \ln 2$, the entropy of full-contrast interference between the two classical branches compatible with one observed momentum. In phase space this interference is the chord structure of the Wigner function. Replacing the classical distribution by the Wigner function gives back the von Neumann entropy identically; replacing it by the Husimi function gives an intermediate value with a closed-form dependence on the coherent-state width, $(1 - \ln 2) - D(\sigma)$, $D \simeq 0.080/\sigma$. No phase-space route reproduces the entanglement entropy, constant term included, without one conventional parameter: a cell for the oscillators, a coherent-state width for the Husimi function, or, when both are fixed as here, an explicit correction $1 - \ln 2$.

At weak coupling the entropy is controlled by the $O(\lambda^2)$ normalization of the perturbed state; at the separatrix both entropies peak; in deep libration the entropy is $\frac{1}{4}\ln(\lambda I/\hbar^2) + \frac{1}{2}\ln(\pi e/\sqrt{2})$ up to an anharmonic correction verified to three digits.

In the bosonic Josephson junction the Schmidt weights

are the atom-number distribution of one well at fixed total number, so the entropies compared here are, in principle, number-statistics measurements, and the separatrix is the onset of running-phase motion [7, 9]. Natural extensions are unequal moments of inertia (exchange symmetry broken, J conserved), $N > 2$ rotors, where the relative motion becomes non-integrable, and quenches of λ across the separatrix.

ACKNOWLEDGMENTS

[To be completed.]

Appendix A: Perturbation theory

Write $H = H_0 + V$ and expand $\left|\widetilde{\Psi}\right\rangle = |0\rangle + \left|\psi^{(1)}\right\rangle + \left|\psi^{(2)}\right\rangle + \dots$ with intermediate normalization $\langle 0|\psi^{(k)}\rangle = 0$. Then $E^{(1)} = V_{00}$, $\left|\psi^{(1)}\right\rangle = \sum_{k\neq 0} V_{k0}\,|k\rangle\,/(E_0 - E_k)$, $E^{(2)} = \sum_{k\neq 0} |V_{k0}|^2/(E_0 - E_k)$, and

$$\left|\psi^{(2)}\right\rangle = \sum_{k,l\neq 0} \frac{V_{kl}V_{l0}\,|k\rangle}{(E_0 - E_k)(E_0 - E_l)} - \sum_{k\neq 0} \frac{V_{00}V_{k0}\,|k\rangle}{(E_0 - E_k)^2}. \tag{A1}$$

From Eq. (3), $V_{00} = 0$ and the only non-zero V_{k0} are $V_{\pm,0} = \lambda/2$ for $|\pm\rangle = |m_1 \pm 1, m_2 \mp 1\rangle$, with $E_0 - E_\pm = \mp\hbar^2(n \pm 1)/I$ and, for $|\pm\pm\rangle = |m_1 \pm 2, m_2 \mp 2\rangle$, $E_0 - E_{\pm\pm} = \mp 2\hbar^2(n \pm 2)/I$. These give Eq. (10), the amplitudes $c_\pm$, and $c_{\pm\pm}$ quoted in Sec. III; the state $|0\rangle$ itself is excluded from the second-order sum. The normalized state is $\mathcal{N}\left|\widetilde{\Psi}\right\rangle$ with $\mathcal{N}^{-2} = 1 + \varepsilon_+ + \varepsilon_- + O(\lambda^4)$.

To second order, $\rho = |\Psi\rangle\langle\Psi|$ contains thirteen entries: the $|0\rangle\langle 0|$ element $1 - \varepsilon_+ - \varepsilon_-$, four first-order coherences $c_\pm(|\pm\rangle\langle 0| + \text{h.c.})$, the populations $\varepsilon_\pm$, the coherence $c_+c_-(|+\rangle\langle -| + \text{h.c.})$, and four entries $c_{\pm\pm}(|\pm\pm\rangle\langle 0| + \text{h.c.})$. One checks $\operatorname{Tr}\rho = 1$ and $\operatorname{Tr}\rho^2 = (1-\varepsilon)^2 + 2\varepsilon + O(\lambda^4) = 1$ with $\varepsilon = \varepsilon_+ + \varepsilon_-$; both fail if the normalization term is omitted. Under the partial trace, $\operatorname{Tr}_2 |a,b\rangle\langle c,d| = \delta_{bd}\,|a\rangle\langle c|$, every coherence connects different rotor-2 labels and drops out, leaving Eq. (12); the $O(\lambda^3)$ terms of ρ are also off-diagonal in m_2, so the error is $O(\lambda^4)$.

Appendix B: Integrals

Band integrals. With $E_{\rm rel} - p^2/I = \lambda\cos u$, $B = \frac{\sqrt{I}}{2}\int \mathrm{d}u\,(E_{\rm rel} - \lambda\cos u)^{-1/2}$. In rotation, $u = \pi - 2t$ gives $E_{\rm rel} - \lambda\cos u = (E_{\rm rel} + \lambda)(1 - k^2\sin^2 t)$ with $k^2 = 2\lambda/(E_{\rm rel} + \lambda)$. In libration, $v = \pi - u$ and $\sin(v/2) = \kappa\sin t$ with $\kappa^2 = (E_{\rm rel} + \lambda)/2\lambda$ give $E_{\rm rel} + \lambda\cos v = (E_{\rm rel} + \lambda)\cos^2 t$ and $\mathrm{d}v = 2\kappa\cos t\,\mathrm{d}t/(1 - \kappa^2\sin^2 t)^{1/2}$. Both lead to Eq. (18), and both parametrize the orbit uniformly in time, $\mathrm{d}t_{\rm orbit} \propto \mathrm{d}t/(1 - k^2\sin^2 t)^{1/2}$, which we use to sample the classical density.

Arcsine entropy and phase average. From $\int_0^\pi \ln\sin u\,\mathrm{d}u = -\pi\ln 2$ and $j = \gamma\cos\beta$ (so $w\,\mathrm{d}j = \mathrm{d}\beta/\pi$), $-\int w\ln w = \ln\pi + \ln\gamma - \ln 2$. With $1 + \cos\Phi = 2\cos^2(\Phi/2)$, $\langle\ln(1 + \cos\Phi)\rangle = -\ln 2$, and integrating by parts, $\langle\cos\Phi\ln(1 + \cos\Phi)\rangle = \langle\sin^2\Phi/(1 + \cos\Phi)\rangle = 1$; the sum is $1 - \ln 2$.

Contrast function. With $s = \sqrt{1 - A^2}$, $\langle\ln(1 + A\cos\Phi)\rangle = \ln[(1 + s)/2]$ [34], and integrating by parts, $\langle\cos\Phi\ln(1 + A\cos\Phi)\rangle = A\langle\sin^2\Phi/u\rangle$ with $u = 1 + A\cos\Phi$. Using $\sin^2\Phi = (A^2 - 1 + 2u - u^2)/A^2$, $\langle u^{-1}\rangle = 1/s$ and $\langle u\rangle = 1$, this is $(1 - s)/A$, which gives Eq. (32). Alternatively, from $(1 + y)\ln(1 + y) = y + \sum_{k\geq 2}(-1)^k y^k/k(k-1)$ and $\langle\cos^{2n}\Phi\rangle = \binom{2n}{n}4^{-n}$, $f(A) = \sum_{n\geq 1}\binom{2n}{n}A^{2n}/[4^n 2n(2n-1)]$; with $\int_0^\infty e^{-4n\sigma^2\beta^2}\mathrm{d}\beta = \sqrt{\pi}/(4\sigma\sqrt{n})$ this gives Eq. (34).

Appendix C: Deep libration

For $p_k \propto e^{-k^2/2\sigma^2}$ on the integers, Poisson summation gives $Z = \sqrt{2\pi}\sigma[1 + 2\sum_{r\geq 1} e^{-2\pi^2\sigma^2 r^2}]$ and $\langle k^2\rangle = \sigma^2[1 + O(\sigma^2 e^{-2\pi^2\sigma^2})]$, so $S = \ln Z + \langle k^2\rangle/2\sigma^2 = \frac{1}{2}\ln(2\pi e\sigma^2)$ up to exponentially small terms. The virial theorem $2\langle T\rangle = \langle xV'\rangle$ with $V = \frac{1}{2}\mu\omega^2 x^2 - gx^4$ gives $\langle T\rangle = \frac{1}{2}(E - \langle gx^4\rangle)$; to first order $E = \frac{1}{2}\hbar\omega - g\langle x^4\rangle_0$ and $\langle x^4\rangle_0 = 3\hbar^2/4\mu^2\omega^2$, so $\langle p^2\rangle = \mu\hbar\omega/2 - 3g\hbar^2/2\mu\omega^2 = \mu\hbar\omega/2 - \hbar^2/16$. If $p = \phi_\sigma(1 + \epsilon\eta)$ with ϕ_σ the Gaussian of the exact variance, η is orthogonal to 1, k, k^2 and the entropy changes only at $O(\epsilon^2)$. Expanding $\frac{1}{2}\ln[2\pi e(\sigma_0^2 - \frac{1}{16})]$ gives Eq. (25).

Appendix D: Wigner function on the circle

The kernel $\langle\theta + \xi/2|\,\rho\,|\theta - \xi/2\rangle = (2\pi)^{-1}\sum_{mm'}\rho_{mm'}e^{i(m-m')\theta}e^{i(m+m')\xi/2}$ is 4π-periodic in ξ, so its Fourier variable ℓ runs over $\frac{1}{2}\mathbb{Z}$; using $\int_{-2\pi}^{2\pi} e^{is\xi}\mathrm{d}\xi = 4\pi\delta_{s0}$ for $s \in \frac{1}{2}\mathbb{Z}$ gives Eq. (26). For two rotors, $W = (2\pi)^{-2}\sum\rho_{m_1m_2;m_1'm_2'}e^{i(m_1 - m_1')\theta_1 + i(m_2 - m_2')\theta_2}$ with $m_i + m_i' = 2\ell_i$; the θ_2 integral sets $m_2' = m_2$ and the sum over ℓ_2 sums over m_2, giving W_{ρ_1}. For an eigenstate, $\rho_{m_1m_2;m_1'm_2'} = c_k\bar{c}_{k'}$ with $2\ell_1 = M + k + k'$, $2\ell_2 = M - k - k'$ and phase $(k - k')(\theta_1 - \theta_2)$, which is Eq. (27). For the classical distribution, $\int \mathrm{d}\varphi\,\mathrm{d}p\,\delta(E_{\rm rel} - H_{\rm rel}) = 4B$ and $\mathcal{C} = 1/8\pi B$.

Appendix E: Numerical methods and checks

We use $\hbar = I = 1$ and the sector $M = 0$. The pendulum is diagonalized in the parity sectors with bases $\{\chi_0, (\chi_k + \chi_{-k})/\sqrt{2}\}$ (where $\langle\chi_0|\,h\,(\chi_1 + \chi_{-1})/\sqrt{2}\rangle = \lambda/\sqrt{2}$) and $\{(\chi_k - \chi_{-k})/\sqrt{2}\}$, $\chi_k = e^{ik\varphi}/\sqrt{2\pi}$, with truncation $|k| \leq 8\sqrt{\lambda} + 150$. Classical cell weights use adaptive quadrature with the band-edge singularities supplied; Bessel sums run over $|j| \leq \gamma + 40\gamma^{1/3} + 50$. Wehrl

entropies of discrete weights are computed by convolution on $\geq 10/\sigma$ points per cell; those of the classical density by sampling the orbit uniformly in time (2×10^5 points), histogramming at resolution 0.01 and convolving. Table III lists the checks quoted in the text.

TABLE III. Analytic results against exact numerics ($\hbar = I = 1$). Perturbative checks at $n = 7$ ($m_1 = 9, m_2 = 2$), $\lambda = 0.02$; Mathieu checks at $q = 14.6$; Wigner checks at $\lambda = 200$, $E_{\rm rel}/\lambda = 1.54$.

quantity	analytic	numerical
c_+	-0.0012500	-0.0012500
c_-	0.0016667	0.0016667
c_{++}	6.944444×10^{-7}	6.944456×10^{-7}
c_{--}	1.666667×10^{-6}	1.666669×10^{-6}
$E^{(2)}$, Eq. (10)	4.166667×10^{-6}	4.166668×10^{-6}
S, $n=0$, Eq. (13)	0.0020420	0.0020411
S, $n=30$, $\lambda=0.05$	2.1084×10^{-5}	2.1146×10^{-5}
$M=0$ lowest level vs $a_0/4$	-5.4542890	-5.4542890
$M=1$ lowest level vs $b_1/4$	-5.4542787	-5.4542787
B, $E_{\rm rel}=300$, $\lambda=100$	0.0927037	0.0927037
B, $E_{\rm rel}=60$, $\lambda=100$	0.1596085	0.1596085
p_0, $n=400$, $\gamma=5$ vs $J_0^2(5)$	0.0315406	0.0314917
$S_{\rm cl}^{\rm bin}-S_q$, $\gamma=10^6$ vs $1-\ln 2$	0.3068528	0.2978385
S_0, $\lambda=10240$, Eq. (25)	3.2067189	3.2067169
$\sum_\ell \int W_{\rm rel}$	1	1.000000000000
$\max_{\ell\in\mathbb{Z}} \lvert\int W_{\rm rel} - p_\ell\rvert$	0	3.8×10^{-17}
$\max_{\ell\in\mathbb{Z}+\frac12} \lvert\int W_{\rm rel}\rvert$	0	3.0×10^{-17}
$\sigma D(\sigma)$ at $\sigma=8$ vs c_1	0.080171	0.080171
$S_W - S_q$, $\gamma=10^4$, $\sigma=1$	0.2267	0.2274

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